

MIDRAND GRAND CENTRAL, South Africa

WMO: 682650

Lat: 25.986S

Lon: 28.140E

Elev: 5318

StdP: 12.08

Time Zone: 2.00 (E02)

Period: 05-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.1	37.3	6.4	9.3	56.3	9.0	10.6	56.0	19.3	43.4	17.6	47.5	11.5	170	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.3	84.5	58.0	82.6	57.7	80.7	58.1	64.7	74.5	64.0	73.7	63.0	72.4	8.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	101.8	67.4	61.0	97.7	66.9	60.5	96.0	66.8	32.8	74.4	32.1	74.0	31.5	73.2	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 19.1	16.8	15.0	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	26.2	66.8	1.9	1.2	24.9	67.7	23.8	68.4	22.7	69.1	21.3	70.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	67.8	68.3	65.0	60.2	58.6	52.8	52.2	55.1	63.0	65.5	65.7	66.8
	DBStd	7.46	3.90	3.71	4.84	4.94	5.10	4.76	4.51	6.36	5.90	5.19	5.22	4.04
	HDD50	90	0	0	0	2	7	27	30	23	1	0	0	0
	HDD65	1764	19	12	58	155	203	367	397	312	104	57	52	28
	CDD50	4359	551	511	464	306	274	111	98	180	391	481	471	521
	CDD65	558	105	104	57	9	5	0	0	4	44	73	73	84
	CDH74	3824	536	596	300	52	27	0	0	33	423	744	592	521
	CDH80	578	80	91	30	0	0	0	0	0	43	158	111	65
(14) Wind	WSAvg	7.4	7.4	7.2	6.8	6.3	6.1	6.9	6.7	7.9	8.2	9.0	8.7	8.0
(15) Precipitation	PrecAvg	27.2	4.9	4.1	3.8	1.5	0.6	0.3	0.1	0.2	0.6	2.4	3.8	5.0
	PrecMax	44.2	9.1	12.2	12.9	4.6	3.9	1.3	0.5	1.2	2.8	6.6	7.8	7.5
	PrecMin	17.6	1.8	0.9	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.7	1.7
	PrecStd	6.6	2.0	2.6	2.7	1.1	0.9	0.4	0.1	0.3	0.7	1.6	1.7	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.7	86.1	83.8	78.4	77.1	70.2	70.0	77.4	83.6	86.3	86.2	84.5
		MCWB	59.6	60.3	59.5	55.2	52.3	48.5	46.5	50.8	52.1	56.3	58.9	59.9
	2%	DB	82.3	82.8	80.2	75.3	73.7	67.9	67.7	73.6	80.9	84.2	82.8	82.1
		MCWB	60.7	59.7	59.0	54.8	50.5	46.9	46.0	50.1	52.1	55.0	58.0	59.6
	5%	DB	79.1	80.5	77.3	73.2	71.5	65.8	64.7	70.2	78.9	80.9	80.4	79.1
		MCWB	61.7	60.5	58.8	55.0	50.6	46.4	45.4	48.5	52.2	55.6	58.7	60.1
	10%	DB	76.9	77.4	75.0	70.2	69.5	62.8	62.6	67.8	76.6	78.5	77.2	77.0
		MCWB	62.2	60.6	59.3	54.3	50.6	45.7	45.1	47.3	51.4	55.3	57.4	59.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	65.5	64.5	61.8	58.2	51.6	50.4	53.6	57.0	61.5	63.3	65.4
		MCDB	75.4	74.4	72.4	70.4	67.5	63.9	61.7	70.7	73.0	74.6	76.6	75.8
	2%	WB	64.8	64.4	63.0	60.0	55.5	50.1	48.7	52.1	55.6	60.0	62.0	64.1
		MCDB	74.4	74.7	71.4	67.7	64.8	61.7	60.9	68.1	71.0	73.8	74.5	74.0
	5%	WB	64.3	63.2	62.1	58.1	53.8	49.0	47.2	50.6	54.4	59.0	61.0	62.9
		MCDB	73.7	73.5	70.8	66.4	64.8	60.5	60.5	64.9	70.9	72.9	73.6	72.4
	10%	WB	63.3	62.4	61.3	56.7	52.5	47.7	46.0	49.3	53.1	57.9	60.0	62.0
		MCDB	72.1	72.8	70.2	65.7	64.1	59.1	59.6	63.2	70.7	71.3	71.8	71.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	19.3	18.1	18.6	19.4	19.6	20.2	21.5	23.3	22.8	20.0	18.9
		MCDBR	20.8	22.3	21.2	20.9	20.7	20.7	21.4	22.3	24.3	25.3	23.0	21.8
	5% WB	MCWBR	7.4	9.1	7.4	7.8	8.7	9.5	9.6	9.4	9.7	8.8	8.3	8.0
		MCDWR	17.0	18.8	16.9	17.4	18.1	17.8	19.8	21.7	23.4	22.3	20.2	18.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.384	0.386	0.375	0.328	0.293	0.275	0.276	0.329	0.369	0.402	0.364	0.371
		taud	2.333	2.326	2.337	2.440	2.518	2.542	2.525	2.368	2.270	2.213	2.359	2.359
	Ebn at Noon	Ebn at Noon	304	298	292	292	291	291	294	288	289	291	308	309
		Edh at Noon	43	43	40	33	28	25	27	35	42	47	42	42
(44) All-Sky Solar Radiation	RadAvg	2103	1991	1765	1512	1384	1256	1364	1607	1913	2051	2136	2153	
	RadStd	220	178	162	134	71	62	35	74	88	140	193	128	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	N/A	-326	+421	+157

Nomenclature: See separate page